

L12: Optimization-based Network Formation Model

The models we examined so far are all probabilistic in nature. In practice, and especially in the technological world, networks are *not* designed randomly. Instead, they are designed to maximize a certain objective function (e.g., minimize cost) subject to one or more constraints. For example, the topology of a computer network may be designed so that it requires few links (routers and their interfaces can be costly) and it offers short paths (and thus small delays).

In social networks, some form of optimization may also be going on "under the surface". For instance, maintaining a friendship requires time and effort – and so we all (maybe subconsciously) try to have a manageable number of social connections, while at the same time we meet our social need for communication.

Even in biological networks, we may have some underlying optimization through evolutionary mechanisms that gradually select the genotypes with the fittest phenotypes.

So, what happens when a network is designed through an optimization process, without any randomness? What kind of network topologies do we get through that process? We will study here only one such optimization model (out of many different formulations, especially in the literature of communication networks).

Suppose that we design a growing communication network. Nodes arrive one at a time on a given "spatial map". For simplicity, let us assume that that map is *an one-by-one square*.

Every new node i connect to one of the existing nodes. If node- i chooses to connect with node- j , the cost of that connection is proportional to $d_{i,j}$, i.e., the Euclidean distance between the two nodes.

This link cost is not our only consideration however. We also want to keep the path length between any pair of nodes short. One way to quantify this objective is the following: Suppose that the very first node represents the "center" of the network – and let h_j be the path length (in number of hops or links) between node- j and the center. If we keep h_j short for all nodes, then the path length between any two nodes will also be short.

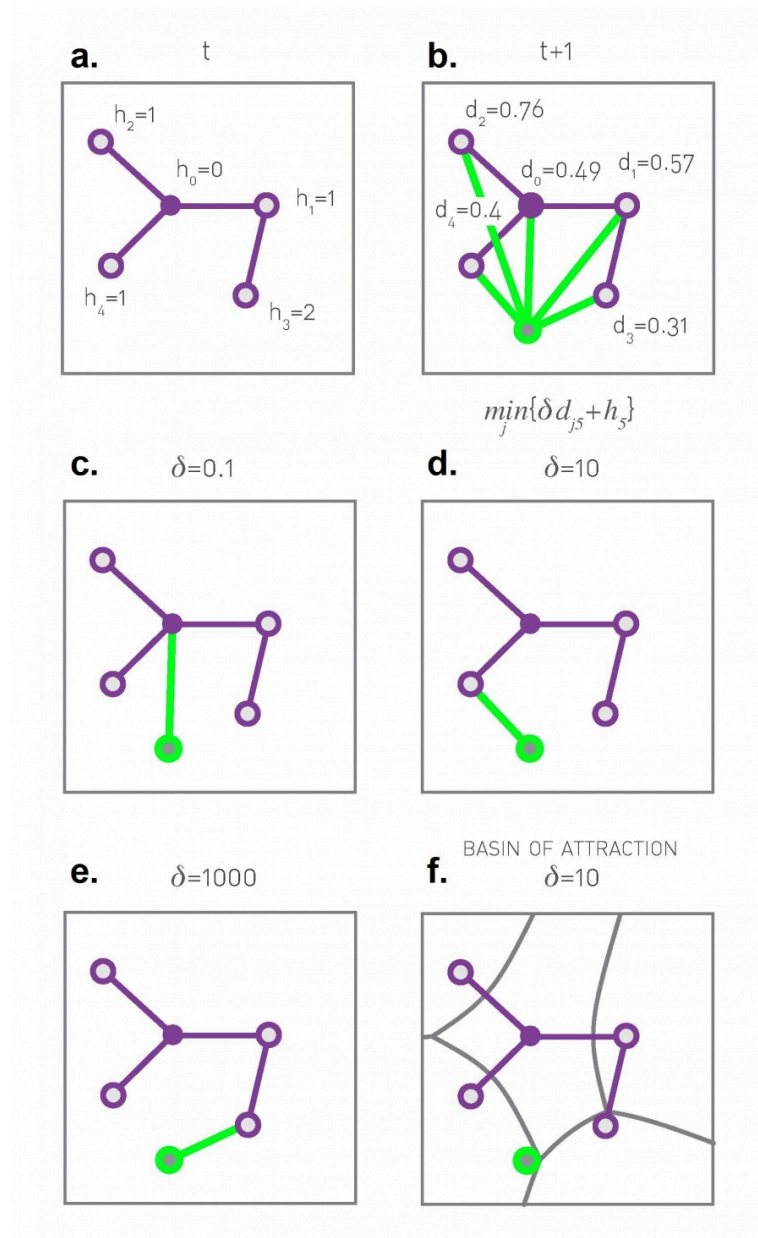
Note that because a new node connects to only one existing node, the resulting topology is a *tree* (and so there is only one path between any two nodes). A tree allows the network to be connected with the minimum number of links ($n-1$ links for n nodes).

So, on one hand, we want to minimize the cost of the new connection, which is proportional to $d_{i,j}$, and on the other hand, we want to connect to a node j with the smallest possible h_j . How can we combine the previous two objectives? Note that they can be conflicting – the node with the smallest h_j may be quite far away from the new node- i .

One approach is to minimize a linear combination of these two metrics. So, when node- i arrives, it connects to the node- j that minimizes the following cost:

$$C_j = \delta d_{i,j} + h_j$$

where δ is a model parameter that determines how much weight we give to the cost of the new connection versus the path length objective.



The visualization (a) shows the value of h for the first five nodes in the network. Plot (b) shows the Euclidean distance between the new node (shown in green) and every other node. The plots (c through e) show the optimal selection for node- j for three different values of δ .

If δ is very low, the new node will prefer to connect directly to the center because the distance-related cost does not matter much.

If δ is very large, the new node will prefer to connect to the nearest existing node in the unit square.

For intermediate values of δ , both objectives matter and the new node connects with the node that optimizes that trade-off, at least heuristically, based on the nodes that are already in the network.

Plot (f) shows that for a given value of δ , and for each node- j , we can identify a “basin of attraction”, i.e., a region of the unit square in which any new node would decide to connect to node- j .

Now that you have some intuition about this model, think before you move to the next page: *what kind of network topology do you expect from this model when δ is very small?* What if δ is very large? And what may happen for intermediate values of δ ?

Food For Thought

Think about other network optimization formulations that present a trade-off between network cost, network efficiency, network reliability, and potentially other properties.